

# BREADCRUMBS — Team Pitch, Demo & Onboarding

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*One doc. Read it once, then don't fumble the pitch, the demo, or the Q&A. Where a line is marked as something to say out loud, say it verbatim — the honest framing is our credibility.*

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## 1. TL;DR — one-liner + what it is

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### The mission (lead with this):

"Our food supply is a black box. BREADCRUMBS makes it traceable — it knows where a food product came from and where it went, so the instant something is wrong, you can follow the crumbs back to the source and forward to every plate at risk."

### The sharp, working proof (what the demo shows):

"The highest-stakes version of traceability is a recall. An FDA recall drops, and in seconds BREADCRUMBS traces the affected lot to exactly which locations received it and what to do — on a box in the back office, data never leaving the building."

### What it is (2 sentences):

BREADCRUMBS is an always-on food-traceability agent. It watches the FDA recall feed and, the moment a lot-coded recall drops, maps that product to the sites that received it (via the supplier links), scores how bad it is, and says what to do — hold the lot, swap the distributor, alert the manager — running entirely on-prem because supply-chain data (suppliers, volumes, margins) is a trade secret that can't be POSTed to a cloud model.

**Language discipline.** Lead with **"food traceability"** / **"trace it back to the source"** — that is the mission. Say **"recall responder,"** never **"outbreak detector"** — that is the honest scope of what works today. Go easy on the word "operator" (it reads corporate); prefer **"food business,"** **"the people who move our food,"** or just name them — a restaurant group, a grocer.

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## 2. Why you should care

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Today an operator finds out about a recall from a distributor email, days later, then hunts through spreadsheets to figure out which locations are hit. That's a phone tree across 12, 50, or 200 kitchens.

**BREADCRUMBS did it in four seconds, on a box in the back office, with the supplier list never leaving the building. That's the difference between a recall and an outbreak.**

The stakes are real money. Restaurant per-recall cost runs a median of \$0.04M–\$1.1M per firm, and the reputational tail is far larger. For grocers, the FMI/GMA study puts average direct recall cost at ~\$10M, with 52% of major recalls exceeding \$10M in total impact.

**The contrast card: 4 seconds vs. 2 days.** Land on it and stop talking.

## 2b. How this maps to real US outbreaks (know these cold)

Judges will ask "would this have caught [outbreak]?" Here is the honest, verified answer — we are relevant to the FDA lot-coded ones and we say plainly where we are not:

| Real US outbreak                                                                      | Relevant?                 | Why                                                                                                                              |
|---------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Jif peanut butter, 2022</b><br>(Salmonella)                                        | <b>YES</b>                | A real FDA Class-I lot-coded recall — exactly our trigger. Verified: 24 real Class-I Smucker records in the feed.                |
| <b>McDonald's / Taylor Farms onions, 2024</b> (E. coli)                               | <b>YES</b>                | FDA-regulated produce, lot-coded recall — we map it to the sites that received the lot.                                          |
| <b>Boar's Head deli meat, 2024</b><br>(Listeria — 10 dead, the deadliest of the year) | <b>NO — and we say so</b> | USDA FSIS jurisdiction (meat/poultry), invisible to the openFDA food feed. Our labeled coverage boundary. v2 adds the FSIS feed. |
| <b>Chipotle, 2015</b> (E. coli)                                                       | <b>NO</b>                 | Restaurant cross-contamination; no recall was ever issued and the ingredient was never identified — nothing to trigger on.       |
| <b>Yuma romaine, 2018</b> (E. coli)                                                   | <b>PARTIAL</b>            | Commodity produce → a public advisory, not a lot-coded recall — no specific lot to trace.                                        |

**The honest headline:** we cover ~100% of FDA lot-coded enforcement recalls, and a minority of the full universe of epidemiological outbreaks. We sit *downstream* of CDC/FDA traceback and act the instant a recall posts. Say **"the fastest path from a recall to action,"** never "we catch outbreaks early." Live sanity check on the real cadence: **NY-distributed Class-I recalls in 2025–26 = 24** — roughly 1–2 genuinely high-severity, NY-relevant events per month. A real alert stream, not a firehose and not zero.

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### 3. What's REAL vs. what's MODELED — our credibility

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We separate the two on purpose. This is the credibility backbone: know which bucket everything is in.

| Element                            | REAL / MODELED                | Source                                                            |
|------------------------------------|-------------------------------|-------------------------------------------------------------------|
| Recall facts                       | REAL                          | openFDA food enforcement — e.g. F-0757-2022                       |
| Distributor recall history         | REAL                          | openFDA firm-level recall counts                                  |
| Site critical violations           | REAL                          | NYC DOHMH inspection records<br>( <code>crit_violations</code> )  |
| Site identities / CAMIS            | REAL                          | DOHMH-geocoded establishments                                     |
| Extruded buildings                 | REAL                          | LIDAR geometry (Sixth Borough engine)                             |
| <b>Supplier → site → lot links</b> | <b>MODELED</b>                | Operator supplier assignments — the FSMA-204 gap itself           |
| Operator 12-site portfolio         | <b>MODELED (illustrative)</b> | Demo portfolio                                                    |
| Narration sentence                 | Generated                     | Local Nemotron, on-box; the decision logic is deterministic Mongo |

Two rules that never bend:

- **The modeled supplier→site link is the point, not a hole.** FSMA 204 is the FDA rule that's *supposed* to make this traceable and doesn't yet. We're modeling the gap the law is trying to close. When a real operator plugs in their real supplier list, the modeled layer disappears.
- **A firm with no name match is labeled "unverified," never "clean."** We never let an absence of data read as safety. Jetro's 0-of-1 is a *verified low record*, not proof of a safer product.

**Coverage boundary, stated not hidden:** Source is **FDA food enforcement only**. USDA FSIS-regulated products — deli meats, poultry, most red meat — are **not** in this feed. The 2024 Boar's Head listeria recall (10 dead) would be invisible to this system. We cover ~100% of FDA lot-coded enforcement recalls and do not claim outbreak detection.

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### 4. The 90-second demo script — beat by beat

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Format: **[ACTION]** = what you do on screen · **"SPOKEN"** = say verbatim · *UPGRADE* = only if live is green. Total budget: 90s. Practice the close.

### BEAT 0 — Cold open (0:00–0:12)

**[ACTION]** deck.gl 3D NYC console already up, idle. Slow free-fly over the operator's 12 extruded site blocks, calm/green.

**"This is a real NYC food operator — 12 sites, their actual buildings. BREADCRUMBS is watching FDA recalls for them right now, and every byte of this is running on this box. No internet."**

HONEST TAG (say it): **"The buildings and inspection data are real; this 12-site portfolio is an illustrative operator we modeled — I'll show you what's real underneath in a second."**

### BEAT 1 — The recall drops (0:12–0:25)

**[ACTION]** Trigger the Dole case. On-screen event pops: **FDA Enforcement — F-0757-2022** .

**"A recall just hit the feed. Dole iceberg lettuce. Listeria. Class I — that's the FDA's most serious, 'reasonable probability of death.' This is a real recall, real number."**

PRIMARY PATH (baked, always works): the F-0757-2022 record is loaded from Mongo and replayed on click — deterministic, never fails.

*UPGRADE (only if live is green): "That fired through a live MongoDB change stream — the moment the record hit the collection, the Watcher agent woke up. Nobody polled anything."*

### BEAT 2 — Sites light red (0:25–0:38)

**[ACTION]** 5 of the 12 blocks flash red. Camera auto-pulls to frame them. Supply-flow arcs from the Dole hub light up to those 5 sites.

**"Five of their twelve sites buy Dole lettuce. Those five just turned red. The other seven are fine — and knowing *which* is the whole game."**

HONEST TAG (say it, don't wait to be asked): **"The supplier-to-site link is the one modeled layer — and that's the point: FSMA 204 is the FDA rule that's *supposed* to make this traceable and doesn't yet. We're modeling the gap the law is trying to close. The 5 sites are real CAMIS locations; the buy-relationship is illustrative."**

### BEAT 3 — Risk scoring (0:38–0:52)

**[ACTION]** Click one red site. Side panel: distributor recall history (**Dole: 205 Class-I**) + that site's **critical food-handling violations** from DOHMH.

"Now it scores the exposure with two real signals. Dole's own recall history — 205 Class-I records, that's real openFDA. And this specific site's critical violations from NYC health inspections — the handling failures, not the letter grade in the window."

**HONEST TAG:** "Both of those are real, verified data. The letter grade is theater; critical violations are what actually predict a handling failure."

#### BEAT 4 — Brief + the swap (0:52–1:05)

**[ACTION]** Briefer panel writes the recommendation. Highlight: **swap Dole → Jetro.**

"It writes the manager a brief and makes a call: stop buying from Dole for this SKU, move to Jetro — zero-for-one on recall record versus Dole's 205. That's a decision, not a dashboard."

**HONEST TAG on inference:** "The numbers you're seeing — the 205, the swap logic — are deterministic MongoDB aggregation, not a language model guessing. The local Nemotron model writes the one plain-English sentence the manager reads. That split is deliberate: the *math* can't hallucinate, and it runs offline."

Firms with no name match are labeled "unverified," never "clean" — say this if pushed on Jetro's 0-of-1.

#### BEAT 5 — Report + alert (1:05–1:18)

**[ACTION]** "Generate report" → formal PDF/report renders. Then "Draft alert" → Telegram message composed to the site manager.

"A formal incident report — for the file, for the health department. And a draft alert to the site manager's phone, over Telegram. Draft, not auto-sent — a human still hits go."

**PRIMARY PATH:** report + Telegram draft are baked and render every time.

**UPGRADE (if live):** "That alert can actually send — the Comms agent is wired to Telegram on this box."

#### BEAT 6 — Why-care close (1:18–1:30)

**[ACTION]** Full-screen the timer/contrast card: **4 seconds vs. 2 days.**

"Today an operator finds out about a recall from a distributor email, days later, then hunts through spreadsheets to figure out which locations are hit. BREADCRUMBS did it in four seconds, on a box in the back office, with the supplier list never leaving the building. That's the difference between a recall and an outbreak."

Land on "back office" and stop. Don't add a sentence.

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## 4b. The agents — 5 roles, in plain English

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BREADCRUMBS runs a small **team of 5 software agents**. Each is a focused role that does its job and hands to the next, and every step is written to an inspectable transcript, so you can prove on stage what actually happened. The honest version, no marketing:

| # | Agent          | What it does, in plain words                                                                                                                                               | Real math, or the AI model?                                                                                                                  |
|---|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Watcher</b> | Sits and watches the FDA recall feed. The moment a new recall lands in the database, it wakes the team and loads the recall's details.                                     | Real — a live MongoDB <i>change stream</i> ; the database pushes it the new record, nothing polls.                                           |
| 2 | <b>Tracer</b>  | Answers "am I hit?" — cross-references the recalled product against the suppliers and locations to find which sites received the affected lot.                             | Real query. The supplier→site link is the one <b>modeled</b> piece (labeled everywhere).                                                     |
| 3 | <b>Risk</b>    | Answers "how bad?" — pulls the distributor's real recall history (Dole: 205 Class-I) and counts how many exposed sites already carry real critical health-code violations. | Real — openFDA + NYC DOHMH. Runs <i>at the same time</i> as the Tracer.                                                                      |
| 4 | <b>Briefer</b> | Writes the one-line, plain-English summary of what's happening and what to do.                                                                                             | <b>The only step that uses the AI model</b> (local Nemotron). It writes ONE sentence; every number comes from the database, never the model. |
| 5 | <b>Comms</b>   | Drafts the alert + a formal hold notice and sends it to the manager's phone over Telegram.                                                                                 | Real send (messaging, not AI); works offline with a mock-phone fallback.                                                                     |

**The one thing to understand about the AI:** only the Briefer talks to the language model, and only to phrase a sentence. Every number — sites exposed, violations, the swap target — is computed by deterministic database queries. **That split is deliberate and it's our best answer to a judge:** the math can't hallucinate, and because the model only writes prose, the whole thing runs offline on the box. Unplug the model and the *decisions are identical* — you just lose the nice sentence. (We are also adding a live "ask the agent" box where local Nemotron answers grounded questions over the same real data — that is the model doing visible, on-box work.)

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## 5. Know your stack — capability / potency / if-a-judge-pushes

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Every teammate should be able to speak any one of these six blocks cold. The rule: **say what the layer really does in this build, then the one honest line if a judge probes**. Confidence comes from not overclaiming.

## 5.1 — Local Nemotron on the GB10 (via vLLM)

`nvidia/Llama-3.1-Nemotron-Nano-4B`, served on the box at `localhost:8000` (OpenAI-compatible), confirmed returning real completions.

- **CAPABILITY:** A 4B instruct model running entirely on-prem, generating the human-readable briefing sentence a Comms agent hands to the operator.
- **POTENCY:** Zero network egress — the operator's supplier list never leaves the building — which is the whole reason a food business could actually deploy this.
- **IF A JUDGE PUSHES ("is the AI making the decision?"):** "No — and that's deliberate. The decisions are deterministic MongoDB aggregation; the model only narrates the result into plain English. We keep judgment in code you can audit, and use the LLM for the one thing it's good at offline: writing the sentence. Right now the inference is thin — narration, not reasoning — and we'd own that in the roadmap."

## 5.2 — MongoDB (four uses)

The system of record and the compute engine. Decisions live here, not in the model.

- **CAPABILITY:** One store doing geo, joins, streaming, and persistent agent state: (a) `2dsphere $near` to find operator sites near a risk; (b) `aggregation $lookup / $group` to join recalls → suppliers → sites and roll up exposure; (c) `change-stream Watcher` that fires when a new recall document lands; (d) `agent_memory` that survives restart and dedupes repeat alerts.
- **POTENCY:** The exposure math (which sites turn red, the swap recommendation) is a real deterministic aggregation over real recall + real site data — reproducible, no model in the loop.
- **IF A JUDGE PUSHES ("which of the four are actually live?"):** "Geo `$near`, the `$lookup` aggregation, and `agent_memory` (restart-persistent, deduped) are live and drive the demo. The change-stream Watcher is built and observes inserts, but it isn't yet wired to auto-trigger the full `run()` pipeline — today we trigger the run explicitly. That's the honest edge."

## 5.3 — The Agent Swarm: Watcher → Tracer → Risk → Briefer → Comms

Five deterministic stages; the LLM only touches the last.

- **CAPABILITY:** Each agent computes one concrete thing:
- **Watcher** — detects a new recall document (openFDA enforcement record).
- **Tracer** — resolves that recall's firm/product to the operator's suppliers and the specific sites fed by them (this is where the labeled modeled supplier→site link is applied).

- **Risk** — scores exposed sites using **real** DOHMH critical food-handling violation counts ( `crit_violations` , 0–56, median 4) — not the letter grade — plus the distributor's real recall history.
- **Briefer** — assembles the site list, the action, and the supplier-swap recommendation.
- **Comms** — turns that into the operator-facing message (Nemotron narration).
- **POTENCY**: Four of the five stages are pure deterministic computation over real data; the swarm is an orchestration of auditable steps, not a chain of LLM guesses.
- **IF A JUDGE PUSHES ("how much is the LLM doing?")**: "Exactly one stage — Comms — and only for wording. Everything upstream is Mongo aggregation and scoring. The one modeled input is the supplier→site link inside Tracer, and we label it modeled because it's the FSMA-204 traceability gap itself — the problem we exist to close, not a number we invented and hid."

## 5.4 — deck.gl 9 3D Console

The operator's back-office view; the real-data layers are the point, not the eye-candy.

- **CAPABILITY**: A 3D NYC console rendering: extruded LiDAR buildings, a risk ground-heat layer, supplier hubs, supply-flow arcs from distributor → site, cuisine search-spotlight, and free-fly camera.
- **POTENCY**: The buildings are real LiDAR geometry and the site positions are real geocoded CAMIS locations — when a recall drops, the five genuinely-exposed sites turn red on their actual footprints.
- **IF A JUDGE PUSHES ("is this just a pretty map?")**: "The geometry and site coordinates are real; the supplier→site arcs are the modeled link, same labeled caveat as everywhere else. It's a decision surface — the red sites and the swap arc are the output of the Mongo aggregation, drawn — not a decorative globe."

## 5.5 — Data Assets + Real-vs-Modeled Discipline

Know these numbers cold.

- **CAPABILITY**: All-free, offline-cacheable, verified-this-build feeds — openFDA food enforcement (29,310 recalls; 6,896 undeclared-allergen, the #2 cause; 301 Class-I E. coli), NYC DOHMH (17,204 geocoded restaurants, risk = per-site critical violations), and distributor recall history from real openFDA firm counts (Dole 205 Class-I, Fresh Express 110, Sysco 24, Jetro 0-of-1).
- **POTENCY**: The demo anchor is a real recall — **Dole F-0757-2022**, iceberg lettuce, Listeria, Class I → 5 real exposed sites (real CAMIS) turn red, swap → Jetro (0-of-1) away from Dole (205) — with a verified real twin, **Jif 2022** (Smucker/Salmonella, 24 real Class-I records), to prove it's not cherry-picked.
- **IF A JUDGE PUSHES ("what's real vs. made up?")**: **Real**: recalls, DOHMH critical violations, distributor recall history, buildings. **Modeled and labeled as such**: the supplier→site links and



the illustrative 12-site portfolio. A firm with no name match is labeled **unverified**, never **clean**."

## 5.6 — Why Local Is Load-Bearing / FSMA 204

- **CAPABILITY:** The entire pipeline — Mongo, the swarm, Nemotron narration, the console — runs on the GB10 with no data egress.
- **POTENCY:** An operator's supplier list, volumes, and margins are trade secrets that can't be POSTed to a cloud LLM; **FSMA 204** makes lot-level traceability a legal mandate. Always-on + private is only achievable on a local box — this is why the hackathon's "local business agent" brief is met for real.
- **IF A JUDGE PUSHES ("couldn't you just call GPT for this?"):** "Not for a real operator. The moment you send the supplier graph to a cloud model you've leaked the trade secret the business most wants to protect — and you can't run always-on inside the four walls. Local isn't a constraint we accepted; it's the feature."

## 6. Judge Q&A — the honest kill-shot answers

### Q1. "Unset the API key / pull the Nemotron model — what actually changes?"

"The narration sentence goes generic or templated. Nothing else. Every red site, every risk score, the swap, the report — all deterministic Mongo aggregation. The LLM writes prose; it doesn't make the decision. That's why we're comfortable running it offline — the load-bearing logic can't hallucinate."

### Q2. "Show me the change stream fire live."

If green: "Watch the event log —" drop a record into the collection, Watcher wakes, sites light.  
If not wired live: **"The demo replays a captured recall deterministically so it never fails on stage; the change-stream Watcher is built against MongoDB's `watch()` — I can show you the code path and the agent\_memory that survives a restart and dedupes replays."**  
Never fake a live fire you can't guarantee.

### Q3. "The supplier→site link is modeled — so what's actually real?"

"Real: the recalls (29,310 openFDA enforcement records), the inspection data (17,204 geocoded NYC restaurants with critical-violation counts), the distributor recall histories (Dole 205, Fresh Express 110, Sysco 24), and the buildings. Modeled and labeled as such: which supplier feeds which site, and this illustrative 12-site operator. The modeled layer is exactly the

FSMA-204 traceability gap — the data the law is forcing into existence. When a real operator plugs in their real supplier list, the model layer disappears."

#### Q4. "What about Boar's Head — the deadliest recent one? USDA meat recalls?"

"Blind spot, and we label it. Boar's Head 2024 was USDA FSIS — deli meat, poultry, red meat — and it's invisible to openFDA, which is our feed. We cover ~100% of FDA lot-coded enforcement recalls and none of USDA's. FSIS publishes a parallel recall feed in the same shape; it's a connector, not a redesign. We'd rather tell you the boundary than get caught at it."

#### Q5. "Does this catch outbreaks early?"

"No — and we won't claim it. It's a recall *responder*, not an outbreak *detector*. It fires when the FDA issues a lot-coded recall. It covers 100% of that feed and a minority of real epidemiological outbreaks. What it does, it does in seconds: recall to site-level action. Overclaiming outbreak detection is how these demos lose credibility."

#### Q6. "Why not just a spreadsheet? Why MongoDB?"

"Three things a spreadsheet can't do. Geospatial — 2dsphere `$near` to match recalls to sites by location. The join at query time — `$lookup` / `$group` across recalls, suppliers, sites, violations in one aggregation. And the change stream — the DB itself wakes the agent the instant a record lands, so it's always-on, not a person refreshing a sheet. A spreadsheet is a snapshot; this is a watcher."

#### Q7. "Why does it need to be local? Feels like a constraint you're rationalizing."

"An operator's supplier list, purchase volumes, and margins are trade secrets. No operator will POST that to a cloud LLM to get a recall alert — the alert isn't worth the leak. FSMA 204 makes lot-level traceability a legal mandate, so this data is about to exist in every operator's back office whether they like it or not. Always-on plus private is only possible on a box like the GB10. Local isn't a constraint we're excusing — it's the only shape the customer will accept."

#### Q8. "What's the business past a demo?"

"FSMA 204 compliance is a mandate with a deadline, and mid-size operators — regional chains, ghost-kitchen groups, distributors — have no tooling for it. This is an on-prem appliance: one box, their data stays theirs, priced per site or per location. The wedge is recall response; the moat is that once their real supplier graph lives in it, it becomes the traceability

system of record the law requires. The USDA connector and a distributor-portal ingest are the next two features, not the next two pivots."

## 7. Business potential — from recall-responder to food-traceability platform

### The wedge → platform arc

**Today (real, narrow, working):** an on-prem responder. An FDA lot-coded recall drops, BREADCRUMBS tells a NYC operator which of their sites touched the lot and what to do, in seconds, on a box in the back office. Demo anchor: Dole F-0757-2022 lights up 5 real exposed sites and recommends swapping distributor away from Dole (205 Class-I records) toward Jetro (0-of-1). This is a complete product. It covers ~100% of its actual feed.

**The hinge is a regulation, and it writes our data for us.** The supplier→site links are *modeled* today — that gap is the FSMA-204 problem itself. FSMA 204 (the FDA Food Traceability Rule) makes lot-level traceability a legal recordkeeping mandate for anyone who manufactures, processes, packs, or holds foods on the Food Traceability List. Mandatory compliance is **July 20, 2028** (extended from January 2026 by the Continuing Appropriations Act of 2026). To comply, the operator must generate traceability lot codes and key data elements at each critical tracking event. Those records are exactly the missing edges.

**Tomorrow (the platform):** feed the operator's own FSMA-204 lot/supplier records into the same MongoDB engine, and the *modeled* supplier→site links become *real* provenance graphs. No new algorithm — the same `$lookup` / `$group` aggregation, now running on real edges. The regulation manufactures our ground truth, on hardware the operator already owns. That single substitution opens three adjacent products off the same engine:

- **Continuous supplier scorecarding** — we already carry real openFDA firm recall histories (Dole 205, Fresh Express 110, Sysco 24). Turn the one-time swap into a live score per supplier.
- **Provenance / origin** — "farm or port → your kitchen," lot-traced, once the real KDEs are flowing.
- **General food-awareness** — allergen radar across the 6,896 undeclared-allergen recalls (the #2 recall cause, ~15x E. coli volume) and origin-country risk via UN Comtrade import data.

Honest boundary, up front: this is FDA-only. USDA FSIS (deli meat, poultry, most red meat — Boar's Head 2024, 10 dead) is a labeled blind spot on the feed, not something the roadmap quietly claims to solve.

### Who pays, and why

- **Lead — multi-site restaurant groups.** A recall drops and they don't know which of their 12, 50, or 200 kitchens received the lot; today that's a phone tree. They pay for site-level action in seconds — and they take it on-prem because, in the ops manager's own words, "**my sourcing and my margins never upload anywhere.**" Restaurant per-recall cost runs a median of \$0.04M–\$1.1M per firm.
- **Grocers / retail — shelf-pull defense.** The pain is *over-broad* recalls: pulling a SKU nationwide when 3 lots at 8 stores were affected. Lot-level provenance narrows the pull. FMI/GMA puts average direct recall cost at ~\$10M, 52% of major recalls exceeding \$10M.
- **Distributors / wholesalers** (Baldor, US Foods, Jetro). Under FSMA 204 they sit on the node with the most critical tracking events, so the mandate lands hardest here. They pay for turnkey KDE capture plus an instant downstream-exposure map.
- **Ghost / cloud kitchens.** Thin margins, many virtual brands, high supplier churn, no food-safety staff. They pay for an always-on watch they don't have to staff.

## Why on-prem is the moat, not a constraint

Two forces point at the same box: FSMA 204 makes the records a legal artifact the operator must keep anyway (by July 2028); and the sourcing list, volumes, and margins in those records are trade secrets. A cloud SaaS asks the operator to upload precisely the data that *is* the trade secret. On-prem resolves the contradiction — the GB10 runs always-on local inference (Nemotron via vLLM) plus MongoDB geo/aggregation entirely offline. The decision logic is deterministic Mongo, not a cloud model; the local LLM only narrates. That's the hackathon's "local business agent" brief satisfied for real.

## TAM one-liner + the 12-month "so what"

**TAM one-liner:** the food-traceability technology market is forecast at roughly **\$30B by 2030** (Grand View, 8.8% CAGR; other houses put 2030 anywhere from ~\$10B to ~\$44B). The reachable slice is narrower: **every US multi-site food operator and distributor facing a hard FSMA-204 deadline in July 2028** — a regulator-created buying window.

**The next real milestone (12 months): ingest one operator's actual purchase records.** Replace the illustrative 12-site portfolio with a real operator's real sites and real supplier POs / invoices. That one step converts the modeled edges into real provenance and proves the platform hinge on live data. No hockey stick: one operator, their real records, on their own box. Same engine the whole way up.

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## 8. The formal FDA-aligned report + alert — what the system outputs

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### Why a formal, FDA-aligned format matters

A recall-response document an operator can put in front of a health inspector, an insurer, or a plaintiff's attorney has to read like the regulatory record it's derived from.

- **Trust / adoption.** Food-safety managers already work from FDA Enforcement Reports and 21 CFR Part 7 recall communications. Mapping our output onto those exact fields means the operator recognizes it on sight.
- **Legal posture.** Under 21 CFR 7.49, a recall communication must let the recipient identify the product, cease its use, and act. A dated, on-prem-generated report showing what the operator knew, when, and what action was directed is contemporaneous evidence of a good-faith response. FSMA 204 makes lot-level traceability a legal duty, so a lot-code-anchored exposure record is on the compliance path.
- **Honesty as a feature.** The document separates REAL (openFDA recall facts, DOHMH violations, distributor recall history) from MODELED (the supplier→site links — the FSMA-204 gap itself). Labeling the modeled inference tells the operator exactly which line to verify against their own receiving records before they act.

PDF report template — "Recall Exposure & Action Report"

Worked with the verified demo anchor **Dole F-0757-2022**. Fields tagged `[openFDA]` are populated verbatim from the live FDA food enforcement record; `[BREADCRUMBS]` are computed on-box; `[MODELED]` are inference, not source data.

■ CLASS I RECALL – SERIOUS HEALTH HAZARD ■

Reasonable probability of serious adverse health consequences or death (21 CFR 7.3). ACT ON THIS TODAY.

RECALL EXPOSURE & ACTION REPORT

|                 |                                                                                |
|-----------------|--------------------------------------------------------------------------------|
| Notice ID       | BC-2026-0822-0417 <code>[BREADCRUMBS]</code>                                   |
| Generated       | 2026-08-22 09:14 ET — on-prem (GB10), offline <code>[BREADCRUMBS]</code>       |
| Operator        | [Operator legal name] — 12-site NYC portfolio <code>[MODELED portfolio]</code> |
| Prepared for    | [Food Safety Manager / person responsible]                                     |
| Source event    | FDA Food Enforcement Report, recall <b>F-0757-2022</b>                         |
| Exposure status | <b>5 of 12 sites EXPOSED</b> — action required <code>[BREADCRUMBS]</code>      |

**A. FDA RECALL FACTS** (*mapped one-to-one to openFDA `food/enforcement` fields — the regulatory record*)

| FDA field                    | Value                                                                |
|------------------------------|----------------------------------------------------------------------|
| recall_number                | F-0757-2022 [openFDA]                                                |
| classification               | Class I [openFDA]                                                    |
| recalling_firm               | Dole Fresh Vegetables, Inc. [openFDA]                                |
| product_description          | Iceberg lettuce, packaged / bulk [openFDA]                           |
| reason_for_recall            | Potential contamination with <i>Listeria monocytogenes</i> [openFDA] |
| code_info (lot / date codes) | [Lot codes and best-by dates as published] [openFDA]                 |
| distribution_pattern         | [States / regions of distribution] [openFDA]                         |
| product_quantity             | [Units / cases recalled] [openFDA]                                   |
| status                       | [Ongoing / Completed / Terminated] [openFDA]                         |
| voluntary_mandated           | Voluntary: Firm Initiated [openFDA]                                  |
| recall_initiation_date       | [YYYY-MM-DD] [openFDA]                                               |
| report_date                  | [YYYY-MM-DD] [openFDA]                                               |
| recalling_firm location      | [City, State] [openFDA]                                              |

*Classification key (21 CFR 7.3):* Class I — reasonable probability of serious adverse health consequences or death. Class II — temporary or medically reversible consequences, or remote probability of serious harm. Class III — use not likely to cause adverse health consequences.

## B. EXPOSURE — which of your sites received this lot

[MODELED] — supplier→site links are inferred, not sourced. This is the FSMA-204 traceability gap: absent a shared lot-level ledger, BREADCRUMBS models which sites a recalled lot reached from the operator's supplier assignments. **Verify each row against your own receiving / invoice records before disposition.** Under FSMA 204 the authoritative answer lives in your Receiving KDEs (traceability lot code, product description, quantity, date, location) — this table is the prompt to pull them.

| Site              | CAMIS    | Supplier link → this lot            | Received (modeled) | Verify against          |
|-------------------|----------|-------------------------------------|--------------------|-------------------------|
| Site 03 — Midtown | 41xxxxxx | Dole → iceberg, lot in<br>code_info | Yes                | Receiving log / invoice |
| Site 05 — LES     | 50xxxxxx | Dole → iceberg                      | Yes                | Receiving log / invoice |

| Site                   | CAMIS    | Supplier link → this lot | Received (modeled) | Verify against          |
|------------------------|----------|--------------------------|--------------------|-------------------------|
| Site 07 — Williamsburg | 41xxxxxx | Dole → iceberg           | Yes                | Receiving log / invoice |
| Site 09 — LIC          | 50xxxxxx | Dole → iceberg           | Yes                | Receiving log / invoice |
| Site 11 — Harlem       | 41xxxxxx | Dole → iceberg           | Yes                | Receiving log / invoice |

7 of 12 sites: no modeled receipt of this lot — no action, but retain this notice.

### C. RISK — how bad, per exposed site (two independent, REAL signals)

Distributor recall history (REAL — openFDA firm counts):

| Firm                         | Class-I recall records | Read                               |
|------------------------------|------------------------|------------------------------------|
| <b>Dole</b> (recalling firm) | <b>205</b>             | Extensive Class-I recall history   |
| Fresh Express                | 110                    | High                               |
| Sysco                        | 24                     | Moderate                           |
| <b>Jetro</b> (proposed swap) | <b>0 of 1</b>          | Low record — verified, not "clean" |

Exposed-site hygiene risk (REAL — DOHMH **crit\_violations**, range 0–56, citywide median 4; the operational-hygiene metric, NOT the letter grade):

| Site                   | CAMIS    | crit_violations | vs. median (4) | Handling-risk read         |
|------------------------|----------|-----------------|----------------|----------------------------|
| Site 05 — LES          | 50xxxxxx | 14              | +10            | Elevated — prioritize hold |
| Site 11 — Harlem       | 41xxxxxx | 9               | +5             | Above median               |
| Site 03 — Midtown      | 41xxxxxx | 6               | +2             | Slightly above             |
| Site 07 — Williamsburg | 41xxxxxx | 4               | 0              | At median                  |
| Site 09 — LIC          | 50xxxxxx | 2               | –2             | Below median               |

Site risk compounds product risk: a Class-I *Listeria* lot at a site with 14 critical handling violations is the highest-priority hold, because cross-contamination controls there are already the weakest.

**D. RECOMMENDED ACTION** (deterministic — computed by MongoDB aggregation on-box, not by a model; the local Nemotron writes only the narration line, never the decision)

1. **HOLD & SEGREGATE.** At all 5 exposed sites, immediately quarantine iceberg lettuce matching the F-0757-2022 lot/date codes in `code_info` . Do not use, serve, or further distribute (21 CFR 7.49). Prioritize **Site 05 (LES, 14 crit violations)** and **Site 11 (Harlem, 9)**.
2. **VERIFY.** Match on-hand lot codes to the recall's `code_info` . Pull each site's Receiving records (FSMA-204 KDEs) to confirm or clear the modeled exposure.
3. **DISPOSITION.** Destroy or return per the recalling firm's instructions; document lot, quantity, site, and date.
4. **SWAP — sourcing recommendation:** move iceberg sourcing from **Dole (205 Class-I recall records)** to **Jetro (0 of 1 record)**.
  - *Honest rationale:* Jetro's low count is a **verified low record**, not proof of a safer product — it reflects few matched openFDA recall records for that firm, which can also mean lower recall visibility. Treat this as a lower-recall-history alternative and confirm Jetro's own lot-traceability before switching. A risk-history swap, not a safety guarantee.
1. **LOG.** This notice, timestamped, is your contemporaneous record that the recall was received and acted on.

**Narration (local Nemotron, GB10):** *"Five of your twelve sites received Dole iceberg lettuce matching this Class I Listeria recall. Hold and segregate now, starting with your Lower East Side location, which carries the most critical handling violations. Consider shifting iceberg sourcing to Jetro, which has a far lighter recall record than Dole."*

**E. REAL vs. MODELED — provenance** (see the full honesty table in §3; report-specific rows below)

| Element                                 | REAL / MODELED                | Source                                                        |
|-----------------------------------------|-------------------------------|---------------------------------------------------------------|
| Recall facts (§A)                       | <b>REAL</b>                   | openFDA food enforcement — F-0757-2022                        |
| Distributor recall history (§C)         | <b>REAL</b>                   | openFDA firm-level recall counts                              |
| Site critical violations (§C)           | <b>REAL</b>                   | NYC DOHMH inspection records ( <code>crit_violations</code> ) |
| Site identities / CAMIS                 | <b>REAL</b>                   | DOHMH-geocoded establishments                                 |
| <b>Supplier → site → lot links (§B)</b> | <b>MODELED</b>                | Operator supplier assignments — the FSMA-204 gap              |
| Operator 12-site portfolio              | <b>MODELED (illustrative)</b> | Demo portfolio                                                |
| Narration sentence (§D)                 | Generated                     | Local Nemotron, on-box; decision logic is deterministic Mongo |



**Coverage boundary (stated, not hidden):** Source is FDA food enforcement only. USDA FSIS products — deli meats, poultry, most red meat — are not in this feed (the 2024 Boar's Head listeria recall would be invisible). This report covers ~100% of FDA lot-coded enforcement recalls and does not claim outbreak detection.

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Source: U.S. FDA Food Enforcement Report (openFDA), recall F-0757-2022.  
FDA classification per 21 CFR 7.3; recall-communication elements per  
21 CFR 7.49; traceability framing per FSMA §204 Food Traceability Rule.  
Generated on-prem on the operator's GB10 — supplier lists, volumes, and  
site data never left the building. This document is a response record,  
not a substitute for the recalling firm's official instructions.  
BREADCRUMBS · Notice BC-2026-0822-0417 · 2026-08-22 09:14 ET

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### **SMS / Telegram short-form (outbox format)**

Formal, plain, no hype — what / where / how bad / do this / ref. Two to four lines.

#### **Primary (Class I, exposed):**

FDA CLASS I RECALL — action needed today.  
Dole iceberg lettuce (Listeria), recall F-0757-2022. 5 of your 12  
sites received matching lots: Midtown, LES, Williamsburg, LIC, Harlem.  
DO: hold & segregate now; verify lot codes; do not serve.  
Full report + swap recommendation: BC-2026-0822-0417 (PDF).

#### **Compressed 2-line variant (SMS length):**

FDA CLASS I: Dole iceberg lettuce, Listeria (recall F-0757-2022).  
5 of 12 sites exposed — HOLD & segregate now. Full report: BC-2026-0822-0417.

#### **All-clear variant (no exposure):**

FDA CLASS I recall posted: Dole iceberg lettuce, Listeria (F-0757-2022).  
No modeled exposure across your 12 sites. No action; notice retained on file.

### **Backend field map — wireable, not aspirational**

Every field resolves to data the pipeline already holds:

1. **§A FDA facts** → `recall_number`, `classification`, `recalling_firm`, `product_description`, `reason_for_recall`, `code_info`, `distribution_pattern` — verbatim from the openFDA `food/enforcement` document already in Mongo.
2. **§B Exposure table** → `exposed_sites[]` with `{camis, name, supplier_link}`; the "5 of 12" banner is `len(exposed_sites)` over the operator portfolio.
3. **§C Site risk** → `exposed_sites[].crit_violations` (DOHMH), sorted desc to drive hold priority.
4. **§C Distributor history** → `distributor_risk.class1` (Dole=205) and the swap target's `distributor_risk.class1` (Jetro=0-of-1).
5. **§D Swap + narration** → `recommendation.swap_to` (Jetro) drives the sourcing line; the Nemotron call fills only the narration string, so the deterministic action survives with or without the model.

*Uncertain / not independently verified in this build: the exact `code_info`, `distribution_pattern`, `product_quantity`, and dates for F-0757-2022 are shown as live `[openFDA]` fill-ins rather than hard-coded; the operator's 12-site portfolio and all supplier→site links are labeled MODELED.*

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## 9. Division of labor + the six things we NEVER overclaim

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### Division of labor (2–3 presenters)

**DRIVER (hands on keyboard, talks least).** Owns the console. Executes Beats 0–5 cleanly — camera framing, the trigger click, report + Telegram. Speaks only the short action lines if solo; otherwise silent and precise. **Rule: never narrate and drive at once — if something lags, keep clicking, let the Narrator cover.** Pre-loads the Dole case and confirms Mongo is up before you're called.

**NARRATOR (the voice, faces judges).** Delivers every "SPOKEN" line and every HONEST TAG. Owns pacing and the Beat-6 close — has to land "4 seconds vs. 2 days" and stop talking. Practices the honest tags until they sound like confidence, not apology. If a beat glitches, bridges verbally ("while that renders —") and never says "it's broken."

**Q&A LEAD (the credibility; same as Narrator if only 2).** Fields the 8 questions. Owns the honest framing cold — especially Q4 (Boar's Head), Q5 (not an outbreak detector), Q1 (pull the key).

**Discipline: answer the question asked, give the real boundary, stop. Never fill silence with an overclaim.** Knows the real numbers by heart: 29,310 / 6,896 / 205 / 110 / 24 / 0-of-1.

- **If 2 people:** Driver + (Narrator = Q&A Lead).
- **If 3:** split all three; Narrator and Q&A Lead pre-agree a handoff cue so judge questions don't collide.

**Shared pre-flight (all present, 60 sec before stage):** Mongo up · console renders 12 blocks · Dole trigger fires red · report generates · Telegram draft composes · timer card loads. If any live upgrade is red, agree *now* to run baked-only and cut the UPGRADE asides — decided before you walk up, never mid-demo.

## **The six lines we NEVER overclaim**

1. **Recall responder, not outbreak detector.** Say "turns an FDA recall into site-level action in seconds." Never "catches outbreaks early" — we cover ~100% of our feed (FDA lot-coded recalls) and only a minority of real epidemiological outbreaks.
2. **FSIS is a labeled blind spot.** We're FDA-only. USDA FSIS (deli meat, poultry, most red meat) is invisible to us — **Boar's Head 2024 (10 dead)** wouldn't appear. Pre-empt it; don't get caught.
3. **Local inference is currently thin.** The LLM writes one narration sentence; deterministic Mongo makes every decision. Frame as strength, don't hide it.
4. **Change-stream isn't wired to `run()` yet.** The Watcher observes inserts; it does not yet auto-trigger the full pipeline. We trigger runs explicitly today.
5. **Supplier→site links are modeled.** They're the FSMA-204 gap we exist to close, labeled modeled — not real traceability data we're passing off as ground truth.
6. **Telegram is draft-not-send.** The Comms output is prepared, not auto-dispatched to a live operator channel.